

1.-18. (canceled)

19. (new) A system for generating recipient-associated distinguishable versions of audio-video content and identifying a recipient associated with a suspected unauthorized distribution of the audio-video content, the system comprising one or more processors and memory storing instructions that, when executed by the one or more processors, cause the system to:

receive video captured from a plurality of cameras and a structured list of edit instructions comprising, for each of a plurality of director-commanded camera cuts, a first edit entry comprising an out-point time code and a first source-camera identifier identifying a first camera of the plurality of cameras as a first source camera, and a following edit entry comprising an in-point time code and a second source-camera identifier identifying a second camera of the plurality of cameras as a second source camera different from the first source camera;

produce reference audio-video content according to the structured list of edit instructions;

generate one or more mates of the reference audio-video content by, for each of a plurality of selected camera cuts among the director-commanded camera cuts, modifying, in the structured list of edit instructions, the out-point time code of the corresponding first edit entry and the in-point time code of the corresponding following edit entry while retaining the first and second source-camera identifiers and the order of the first and second source cameras, such that, for each selected camera cut:

a transition from the first source camera to the second source camera occurs at a reference camera-switch timing in the reference audio-video content and at a mate camera-switch timing different from the reference camera-switch timing in at least one of the one or more mates; and

during a temporal interval between the reference camera-switch timing and the mate camera-switch timing, the reference audio-video content and the at least one mate contain temporally corresponding frames captured by different ones of the first and second source cameras;

segment an ensemble comprising the reference audio-video content and the one or more mates into chunks;

generate a plurality of manifest files pointing to respective combinations of chunks selected from the ensemble, each respective combination of chunks preserving, for each of the plurality of selected camera cuts, a choice between the reference camera-switch timing and the mate camera-switch timing of that selected camera cut, so that the respective combinations of chunks represent respective combinations of timing choices across the plurality of selected camera cuts;

cause delivery of streamed audio-video content to respective recipients according to respective manifest files of the plurality of manifest files;

store, in a ledger, associations between the respective manifest files and the recipients to which the streamed audio-video content was delivered according to the respective manifest files;
receive the suspected unauthorized distribution;

apply a scene-change detection algorithm to the suspected unauthorized distribution to identify a plurality of time codes of camera cuts in the suspected unauthorized distribution;

build one or more reconstructed manifest files from the plurality of identified time codes, the one or more reconstructed manifest files including a reconstructed manifest file representing a detected combination of camera-switch timings across the plurality of selected camera cuts; and

search the ledger to identify a recipient associated with a delivered manifest file representing a recipient-associated combination of timing choices, wherein, for each of the plurality of selected camera cuts, the recipient-associated combination and the detected combination represented by the reconstructed manifest file include the same one of the reference camera-switch timing and the mate camera-switch timing for that selected camera cut, and wherein, for at least one of the plurality of selected camera cuts, that same one is the mate camera-switch timing different from the reference camera-switch timing.

20. (new) The system of claim 19, wherein one of the plurality of selected camera cuts is a delayed selected camera cut for which the mate camera-switch timing is later than the reference camera-switch timing, and wherein the instructions cause the system to preserve, in the mate containing the mate camera-switch timing for the delayed selected camera cut, a timing of a subsequent camera cut from the reference audio-video content, thereby restoring synchronization between that mate and the reference audio-video content from the subsequent camera cut onward.

21. (new) The system of claim 19, wherein the structured list of edit instructions is an Edit Decision List.

22. (new) The system of claim 19, wherein the video is captured live from diverse viewpoints and the structured list of edit instructions records the director-commanded camera cuts as real-time selections among the plurality of cameras.

23. (new) The system of claim 19, wherein the instructions cause the system to generate a respective mate for each selected camera cut, each respective mate differing from the reference audio-video content in the modified out-point and in-point time codes for only that selected camera cut.

24. (new) The system of claim 19, wherein identifying the plurality of time codes comprises comparing frames of the suspected unauthorized distribution with frames of one or both of the reference audio-video content and the one or more mates using perceptual hashes of the frames.

25. (new) The system of claim 24, wherein the comparing comprises fuzzy matching in which perceptual hashes of groups of frames are compared using sliding windows.

26. (new) The system of claim 19, wherein the chunks are distributed through a content delivery network using adaptive streaming, the plurality of manifest files is tailored to recipient devices or network conditions, and a mixing process integrates chunks of the reference audio-video content with chunks of the one or more mates and progressively assigns respective manifest files to recipients as additional selected camera cuts become available.

27. (new) The system of claim 19, wherein a first manifest file points to a first chunk selected from the reference audio-video content and a second manifest file points to a second chunk selected from one of the one or more mates, the first chunk and the second chunk spanning the same playback interval and having equal playback durations.

28. (new) The system of claim 27, wherein, for one of the selected camera cuts:

the first chunk contains frames from the corresponding first source camera before the reference camera-switch timing and frames from the corresponding second source camera after the reference camera-switch timing; and

the second chunk contains frames from the corresponding first source camera before the mate camera-switch timing and frames from the corresponding second source camera after the mate camera-switch timing.

29. (new) The system of claim 19, wherein the delivered manifest file and the reconstructed manifest file identify the same respective chunk selections at corresponding manifest positions associated with the plurality of selected camera cuts.

30. (new) A method of identifying a recipient associated with a suspected unauthorized distribution of audio-video content, the method comprising:

receiving the suspected unauthorized distribution;

applying a scene-change detection algorithm to the suspected unauthorized distribution to identify a plurality of time codes of camera cuts in the suspected unauthorized distribution;

building one or more reconstructed manifest files from the plurality of identified time codes, the one or more reconstructed manifest files including a reconstructed manifest file representing a detected combination of camera-cut timings across a plurality of camera cuts; and

searching a ledger comprising associations between a plurality of delivered manifest files and respective recipients to identify a recipient associated with a delivered manifest file,

wherein each of the plurality of delivered manifest files identifies a respective combination of chunks selected from an ensemble comprising reference audio-video content and one or more

mates of the reference audio-video content, and wherein, for each of the plurality of camera cuts, the reference audio-video content contains an ordered transition from a respective first source camera to a respective second source camera at a respective reference camera-cut timing, and at least one of the one or more mates contains the same ordered transition at a respective different mate camera-cut timing,

wherein each of the plurality of delivered manifest files represents, across the plurality of camera cuts, a recipient-associated combination of choices between the respective reference camera-cut timings and the respective different mate camera-cut timings, and

wherein, for each of the plurality of camera cuts, the recipient-associated combination represented by the delivered manifest file and the detected combination represented by the reconstructed manifest file include the same one of the respective reference camera-cut timing and the respective different mate camera-cut timing, and wherein, for at least one of the plurality of camera cuts, that same one is the respective different mate camera-cut timing detected in the suspected unauthorized distribution.

31. (new) The method of claim 30, wherein the delivered manifest file and the reconstructed manifest file identify the same respective chunk selections at corresponding manifest positions associated with the plurality of camera cuts.

32. (new) The method of claim 30, wherein the delivered manifest file identifies at least one chunk selected from a mate of the one or more mates, the at least one chunk spans a temporal region of the ensemble containing an ordered transition of one of the plurality of camera cuts, and, during an interval between the respective reference camera-cut timing of that ordered transition and the respective different mate camera-cut timing of that ordered transition in the mate, the reference audio-video content and the mate contain temporally corresponding frames captured by different ones of the respective first and second source cameras.

33. (new) A method of identifying a plurality of recipients whose respective delivered streamed audio-video content contributed respective portions to a suspected unauthorized distribution of audio-video content, the method comprising:

accessing a ledger comprising associations between a plurality of delivered manifest files and respective recipients, each delivered manifest file identifying a recipient-associated sequence of chunk selections from an ensemble comprising reference audio-video content and one or more mates of the reference audio-video content, the recipient-associated sequence of chunk selections representing a recipient-associated sequence of timing choices across a plurality of selected camera cuts, wherein, for each selected camera cut, each timing choice is a choice between a reference camera-switch timing at which the reference audio-video content contains an ordered transition from a respective first source camera to a respective second source camera and a mate camera-

switch timing different from the reference camera-switch timing at which a respective mate of the one or more mates contains the same ordered transition, and wherein, during an interval between the reference camera-switch timing and the mate camera-switch timing, the reference audio-video content and the respective mate contain temporally corresponding frames captured by different ones of the respective first and second source cameras;

receiving the suspected unauthorized distribution;

applying a scene-change detection algorithm to a plurality of temporal portions of the suspected unauthorized distribution to identify respective time codes of camera cuts corresponding to respective ones of the plurality of selected camera cuts;

determining, for each identified time code, whether the corresponding camera cut in the suspected unauthorized distribution occurs at the reference camera-switch timing or the mate camera-switch timing, thereby deriving a detected sequence of timing choices for the plurality of temporal portions;

applying a probabilistic fingerprinting algorithm to the detected sequence and to the recipient-associated sequences represented by the plurality of delivered manifest files; and

identifying, based on an output of the probabilistic fingerprinting algorithm, a plurality of recipients whose delivered streamed audio-video content contributed respective ones of the plurality of temporal portions to the suspected unauthorized distribution.

34. (new) The method of claim 33, wherein the probabilistic fingerprinting algorithm comprises a segmented Tardos fingerprinting algorithm that applies respective Tardos fingerprints to content segments corresponding to the plurality of temporal portions and identifies, for respective temporal portions, respective contributing recipients among the plurality of recipients.

35. (new) A method for generating recipient-associated distinguishable versions of audio-video content and identifying a recipient associated with a suspected unauthorized distribution of the audio-video content, the method comprising:

receiving video captured from a plurality of cameras and a structured list of edit instructions comprising, for each of a plurality of director-commanded camera cuts, a first edit entry comprising an out-point time code and a first source-camera identifier identifying a first camera of the plurality of cameras as a first source camera, and a following edit entry comprising an in-point time code and a second source-camera identifier identifying a second camera of the plurality of cameras as a second source camera different from the first source camera;

producing reference audio-video content according to the structured list of edit instructions;

generating one or more mates of the reference audio-video content by, for each of a plurality of selected camera cuts among the director-commanded camera cuts, modifying, in the

structured list of edit instructions, the out-point time code of the corresponding first edit entry and the in-point time code of the corresponding following edit entry while retaining the first and second source-camera identifiers and the order of the first and second source cameras, such that, for each selected camera cut:

a transition from the first source camera to the second source camera occurs at a reference camera-switch timing in the reference audio-video content and at a mate camera-switch timing different from the reference camera-switch timing in at least one of the one or more mates; and

during a temporal interval between the reference camera-switch timing and the mate camera-switch timing, the reference audio-video content and the at least one mate contain temporally corresponding frames captured by different ones of the first and second source cameras;

segmenting an ensemble comprising the reference audio-video content and the one or more mates into chunks;

generating a plurality of manifest files pointing to respective combinations of chunks selected from the ensemble, each respective combination of chunks preserving, for each of the plurality of selected camera cuts, a choice between the reference camera-switch timing and the mate camera-switch timing of that selected camera cut, so that the respective combinations of chunks represent respective combinations of timing choices across the plurality of selected camera cuts;

delivering streamed audio-video content to respective recipients according to respective manifest files of the plurality of manifest files;

storing, in a ledger, associations between the respective manifest files and the recipients to which the streamed audio-video content was delivered according to the respective manifest files;

receiving the suspected unauthorized distribution;

applying a scene-change detection algorithm to the suspected unauthorized distribution to identify a plurality of time codes of camera cuts in the suspected unauthorized distribution;

building one or more reconstructed manifest files from the plurality of identified time codes, the one or more reconstructed manifest files including a reconstructed manifest file representing a detected combination of camera-switch timings across the plurality of selected camera cuts; and

searching the ledger to identify a recipient associated with a delivered manifest file representing a recipient-associated combination of timing choices, wherein, for each of the plurality of selected camera cuts, the recipient-associated combination and the detected combination represented by the reconstructed manifest file include the same one of the reference camera-switch timing and the mate camera-switch timing for that selected camera cut, and

wherein, for at least one of the plurality of selected camera cuts, that same one is the mate camera-switch timing different from the reference camera-switch timing.

36. (new) The method of claim 35, wherein one of the plurality of selected camera cuts is a delayed selected camera cut for which the mate camera-switch timing is later than the reference camera-switch timing, and wherein generating the one or more mates comprises preserving, in the mate containing the mate camera-switch timing for the delayed selected camera cut, a timing of a subsequent camera cut from the reference audio-video content, thereby restoring synchronization between that mate and the reference audio-video content from the subsequent camera cut onward.

37. (new) The method of claim 35, wherein identifying the plurality of time codes comprises comparing frames of the suspected unauthorized distribution with frames of one or both of the reference audio-video content and the one or more mates using perceptual hashes of the frames.

38. (new) The method of claim 37, wherein the comparing comprises fuzzy matching in which perceptual hashes of groups of frames are compared using sliding windows.